

Vineesh Mattapally

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Objective

Data science professional with strong experience in **Python, SQL, and analytics**, supported by a Master's degree in Data Science & Analytics. Seeking new opportunities in the data science field to apply machine learning, data analysis, and visualization skills to solve real-world problems and deliver data-driven insights.

Education

Master's in Data Science and Analytics

Sept 2023 – Dec 2025

Florida Atlantic University (FAU), Boca Raton, Florida

- GPA: 3.78/4.00
- **Coursework:** Cloud Computing, Data Mining and Machine Learning, Deep Learning, Information Retrieval, Social Media Web Analytics, Conversational AI, Intro to Data Science, Big Data Analytics, Computer Data Security

Bachelor's in Computer Science Engineering

2018 – 2022

Maharishi Institute of Information Technology, Lucknow, India

- GPA: 3.00/4.00
- **Coursework:** Data Structures, Operating Systems, Database Systems, Computer Networks

CAR-147 Approved Basic Training Course

2018 – 2021

(Category: B2 – Avionics)

School for Aircraft Maintenance Engineering (SAME), Greater Noida, India

- 288 hours of practical training on Airbus A320 CEO/NEO and Airbus A321 NEO with IndiGo Airlines (IGI Airport, New Delhi)
- **Covered modules:** Electrical Fundamentals, Digital Techniques, Basic Aerodynamics, Aircraft Structures and Systems, Aviation Legislation, and Propulsion

Experience

Graduate Teaching Assistant, Florida Atlantic University – Boca Raton, FL

Feb 2025 – Present

- **Key Tools:** SQL, Python, Excel, Canvas LMS, Git
- **Courses Supported:** Cloud Computing, Intro to Software Design (Graduate & Undergraduate)
- Facilitated **Python programming labs** and guided students through hands-on coding, data handling, and problem-solving exercises.
- Evaluated **90+ research papers**, providing detailed feedback on methodology, structure, and academic integrity using Turnitin reports.
- Analyzed student performance and assessment datasets to identify trends and improvement areas.
- Cleaned and validated datasets used for grading, reporting, and academic analysis.
- Created structured summaries and reports for instructors and students.
- Conducted lab sessions, discussion groups, and tutoring sessions to reinforce programming, data analysis, and cybersecurity concepts.

Associate Analyst, CAMP Systems Private Limited – Hyderabad, India

Apr 2022 – Dec 2023

- **Key Technologies:** SQL, Excel, Data Validation, Documentation Analysis, Quality/Compliance Reporting
- Analyzed enterprise maintenance datasets (logs, work cards, tags, compliance records) and performed data validation for accuracy and completeness.

- Conducted **data profiling** and identified data quality gaps; recommended process improvements to reduce errors and improve consistency.
- Collaborated with cross-functional stakeholders (quality assurance, regulatory, operations) to understand data needs and provide analytical support.
- Documented data processes and requirements to support repeatable workflows and improved knowledge sharing.

Projects

Restaurant Table Booking System 2023

- **Tech Stack:** SQL, Python, Flask, HTML, CSS
- Built and analyzed relational datasets for reservations and customer activity.
- Used SQL queries to track booking trends, peak usage times, and system performance.
- Cleaned and validated reservation data to ensure accuracy and consistency.

Google Cloud Conversational Agent for Book Querying 2024

- **Tech Stack:** Google Cloud Dialogflow CX, Vertex AI Agent Builder, Google Cloud Storage, DataStore, Phone Gateway, Python
- Built a conversational agent using Dialogflow CX to answer information-based questions from the book *Boynnton Bicycle Railway System*.
- Uploaded and indexed the full book PDF in Google Cloud Storage and DataStore to enable accurate AI-driven responses.
- Integrated a phone gateway to enable voice interaction, improving system accessibility and user engagement.
- Configured intents, entity extraction, response flows, and validation to ensure meaningful and context-aware conversations.

Interactive Car Sales Analytics Dashboard (Power BI) 2025

- **Tech Stack:** Microsoft Power BI, Power Query, DAX, SQL, Excel
- Built an interactive Power BI dashboard to analyze dealership sales performance using YTD, MTD, and YoY metrics.
- Cleaned and transformed large automotive sales datasets (13,000+ rows) by standardizing fields, removing duplicates, and handling missing values.
- Designed interactive visuals including weekly sales trends, regional performance maps, color/model-based distributions, and detailed transaction tables.
- Identified high-performing regions and popular vehicle models, supporting data-driven decisions for inventory and marketing.

Hierarchical Semantic Propagation for Object Detection in Remote Sensing Imagery 2024

- **Tech Stack:** CNNs, Feature Pyramid Networks (FPN), Semantic Segmentation, Object Detection, Remote Sensing Analytics, Deep Learning Literature Review
- Studied and summarized the HSP deep-learning framework for small-object detection in aerial and satellite imagery.
- Analyzed multiscale feature fusion, soft segmentation, and semantic-guided RPN methods for improved detection accuracy.
- Reviewed HBB/OBB prediction performance on datasets such as DOTA, UCAS_AOD, and NWPU VHR-10.
- Compared HSP enhancements against baseline models including Faster R-CNN, R2CNN, and RoI-Transformer.

Robot Dog Autonomous Tracking System 2025

- **Tech Stack:** Python, OOP, Class Design, Battery Simulation, Path Tracking, Loop Logic
- Implemented a Battery class with methods for initialization, consumption, and capacity reporting.

- Designed a PathTracker class to record positions, count steps, and maintain a movement history.
- Integrated both classes into the main simulation loop to create a modular robotic behavior system.
- Automated 5% battery consumption per cycle and logged every movement step using class methods.
- Debugged logical and runtime issues in Python-based simulation.
- Implemented logging and status reporting to aid troubleshooting and monitoring.

Certifications

- **Learning Data Analytics: 1 Foundations** — LinkedIn Learning *Issued Jan 2026*
- **Introduction to Career Skills in Data Analytics** — LinkedIn Learning *Issued Dec 2025*
- **CAR-147 Basic Training Course (Category B2 – Avionics)**, School for Aircraft Maintenance Engineering (SAME), Greater Noida, India
- **CodePath Cybersecurity Course** – hands-on labs in ethical hacking, password cracking, cryptography, and secure system design.

Technical Skills

- **Programming:** Python, SQL, Java
- **Data Science & ML:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Predictive Modeling
- **Data Engineering:** Data Cleaning, Data Transformation
- **Cloud:** AWS (Lambda, S3, Glue), Google Cloud (Dialogflow, Storage, Datastore)
- **Databases:** MySQL, PostgreSQL, SQLite
- **Visualization & BI:** Power BI (dashboards, DAX basics, data modeling), Tableau, Matplotlib, Seaborn
- **Cybersecurity:** Secure System Design, Encryption, John the Ripper, SSH
- **Tools:** Microsoft Excel, Outlook, Git, Ticketing Systems
- **Domains:** Aviation Maintenance Systems, Compliance Regulatory Data
- **DevOps & Architecture:** Docker, Containerization, Microservices Architecture (Foundational)